MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FILTER, LOW PASS, ACTIVE, ANALOG, KEY CODE EEE

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic active low pass filter module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-22097 - Resistors, Variable, Nonwire-Wound (Adjustable Type), General Specification for.

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.3289 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EEE.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 20 contact pins positioned A21 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/218-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.5.9 Potentiometers. Potentiometers selected for use in the module shall conform to the requirements of this document and shall be in accordance with MIL-R-22097, except as modified herein. Alternate potentiometers may be selected provided there is documented evidence that they are proven to be equivalent in all functional respects and that they conform to or exceed the reliability requirements of this document. The potentiometer lead screw shall be accessible from the module fin. Access hole shall be as specified in figure 1. The lead screw adjustment of the potentiometer shall have a minimum of 10 mechanical turns.

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 2.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of ±15.0 ±1.5 Vdc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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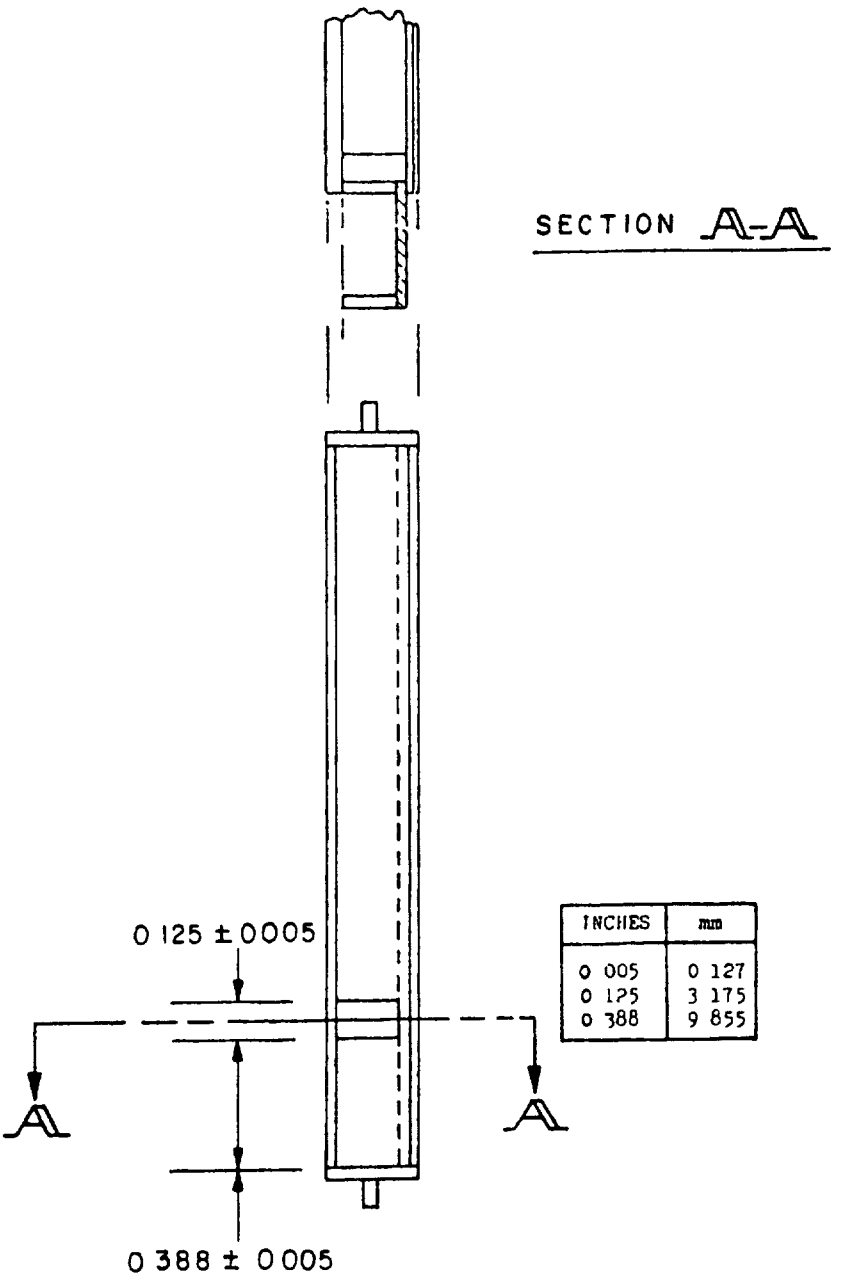
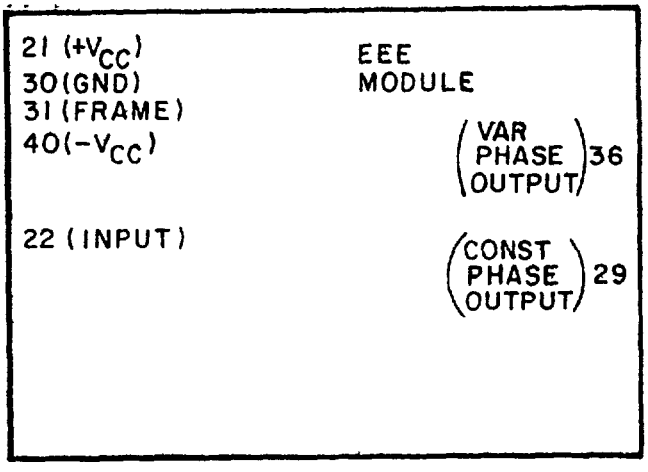


FIGURE 1 Location of potentiometer access hole.

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NOTES:
1. Pins 23 through 28, 32 through 35, and 37 through 39 are not used.
2. Pins 1 through 20, if supplied, are not used (see 6.3).

FIGURE 2. EEE module function diagram, active low pass filter.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1, B2	Output short circuit duration	III	Q	10	---	10	---	10	---	10	---	S
C1	DC offset voltage	III	X	-20	20	-20	20	-20	20	-20	20	mV
C2	DC offset voltage	III	X	-15	15	-15	15	-17	17	-17	17	mV
D1, D2	DC gain	III	Q	0.95	1.05	0 95	1.05	0.95	1 05	0.95	1.05	V
E1, E2	Output voltage swing	IV	S	7.0	---	7.0	---	7.0	---	7.0	---	V
F1, F2	Power supply current	V	Q	1 0	8.5	1.0	8 5	1.0	9 0	1.0	9.0	mA
G1	Gain versus frequency at 9600 Hz	VI	X	-0.5	1.5	-0.5	1 5	-0.5	1.5	-0.5	1.5	dB
G2	Gain versus frequency at 20 to 7000 Hz	VI	X	-2.0	0.5	-2 0	0 5	-2.0	0.5	-2.0	0.5	dB ref G1
G3	Gain versus frequency at 7600 Hz	VI	X	-0.3	0 3	-0.3	0.3	-0.3	0.3	-0.3	0.3	dB ref G1
G4	Gain versus frequency at 9540 Hz	VI	X	-0.25	0.25	-0.25	0.25	-0.25	0.25	-0.25	0 25	dB ref G1
G5	Gain versus frequency check at 9660 Hz	VI	X	-0.25	0.25	-0.25	0 25	-0.25	0.25	-0.25	0.25	dB ref G1
G6	Gain versus frequency check at 11600 Hz	VI	X	-0.3	0.3	-0.3	0 3	-0.3	0.3	-0.3	0.3	dB ref G1
G7	Gain versus frequency check at 15 kHz	VI	X	-0.5	0 5	-0.5	0 5	-0.5	0.5	-0.5	0.5	dB ref G1

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
G8	Frequency at 3 dB down	VI	X	17.5	19.5	17.5	19.5	17.5	19.5	17.5	19.5	kHz
G9	Gain versus frequency check at 50 kHz	VI	X	---	-40	---	-40	---	-40	---	-40	dB ref G1
G10	Gain versus frequency check 16 to 100 kHz	VI	X	← See note 6, figure 9 →								
G11	Gain versus frequency check at 9600 Hz	VI	Q	-0.5	1.5	-0.5	1.5	-0.5	1.5	-0.5	1.5	dB
G12	Gain versus frequency check 20 to 7000 Hz	VI	Q	-2.0	0.5	-2.0	0.5	-2.0	0.5	-2.0	0.5	dB ref G11
G13	Gain versus frequency check at 7600 Hz	VI	Q	-0.3	0.3	-0.3	0.3	-0.3	0.3	-0.3	0.3	dB ref G11
G14	Gain versus frequency check at 9540 Hz	VI	Q	-0.25	0.25	-0.25	0.25	-0.25	0.25	-0.25	0.25	dB ref G11
G15	Gain versus frequency check at 9660 Hz	VI	Q	-0.25	0.25	-0.25	0.25	-0.25	0.25	-0.25	0.25	dB ref G11
G16	Gain versus frequency check at 11600 Hz	VI	Q	-0.3	0.3	-0.3	0.3	-0.3	0.3	-0.3	0.3	dB ref G11
G17	Gain versus frequency check at 15 kHz	VI	Q	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	dB ref G11
G18	Frequency at 3 dB down	VI	Q	17.5	19.5	17.5	19.5	17.5	19.5	17.5	19.5	kHz
G19	Gain versus frequency check at 50 kHz	VI	Q	---	-40	---	-40	---	-40	---	-40	dB ref G11

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
G20	Gain versus frequency check 16 to 100 kHz	VI	Q	← See note 6, figure 9 →								
H1	Phase shift (cw)	VII	X	240	260	240	260	240	260	240	260	Degree
H2	Phase shift (ccw)	VII	X	90	110	90	110	90	110	90	110	Degree
H3	Phase shift	VII	Q	85	100	85	100	85	100	85	100	Degree
H4	Phase shift settability	VII	S	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	Degree
H5	Phase shift setting under pressure	VII	Q	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	Degree
H6	Phase shift average temperature coefficient check	VII	Q	---	---	---	---	0.0	0.05	0.0	0.05	Degree /C
J1, J2	Distortion check	VIII	S	0	2	0	2	0	2	0	2	%
K1, K2	Power supply transient rise time	IX	S	27.5	77.5	27.5	77.5	27.5	77.5	27.5	77.5	µs
L	Continuity	X	S	0	1	0	1	0	1	0	1	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through L shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through L in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through L shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition K shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests C through L shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through L shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/218 supersedes Department of the Navy, Naval Electronic Systems Command Drawing 28687-0102-877. Part number M28787/218-1 supersedes part numbers 0102-877 and M28787/218.

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6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Power dissipation	300 mW maximum (outputs not loaded)
Quiescent power supply current	9 mA maximum each supply
Gain at 50 kHz	-40 dB maximum (reference 9.6 kHz)
Passband flatness	-2.0 dB to +0.50 dB (reference 9.6 kHz, DC to 14.9 kHz)
Power supply voltage	+15 V ±10 percent -15 V ±10 percent
DC gain	Unity (1)
Frequency at -3 dB point	18.5 ±1.0 kHz
Maximum input voltage	±14 V or $\pm V_{CC}$, whichever is less
Output voltage swing	±7 V minimum into 2 kΩ load at 5.0 kHz
Phase shift	110 degrees to 250 degrees, adjustable
Distortion	2 percent maximum at 9.6 kHz and 4 V rms output

6.5 Module function. The module consists of a unity gain 18.5 kHz low pass filter with adjustable phase shift.

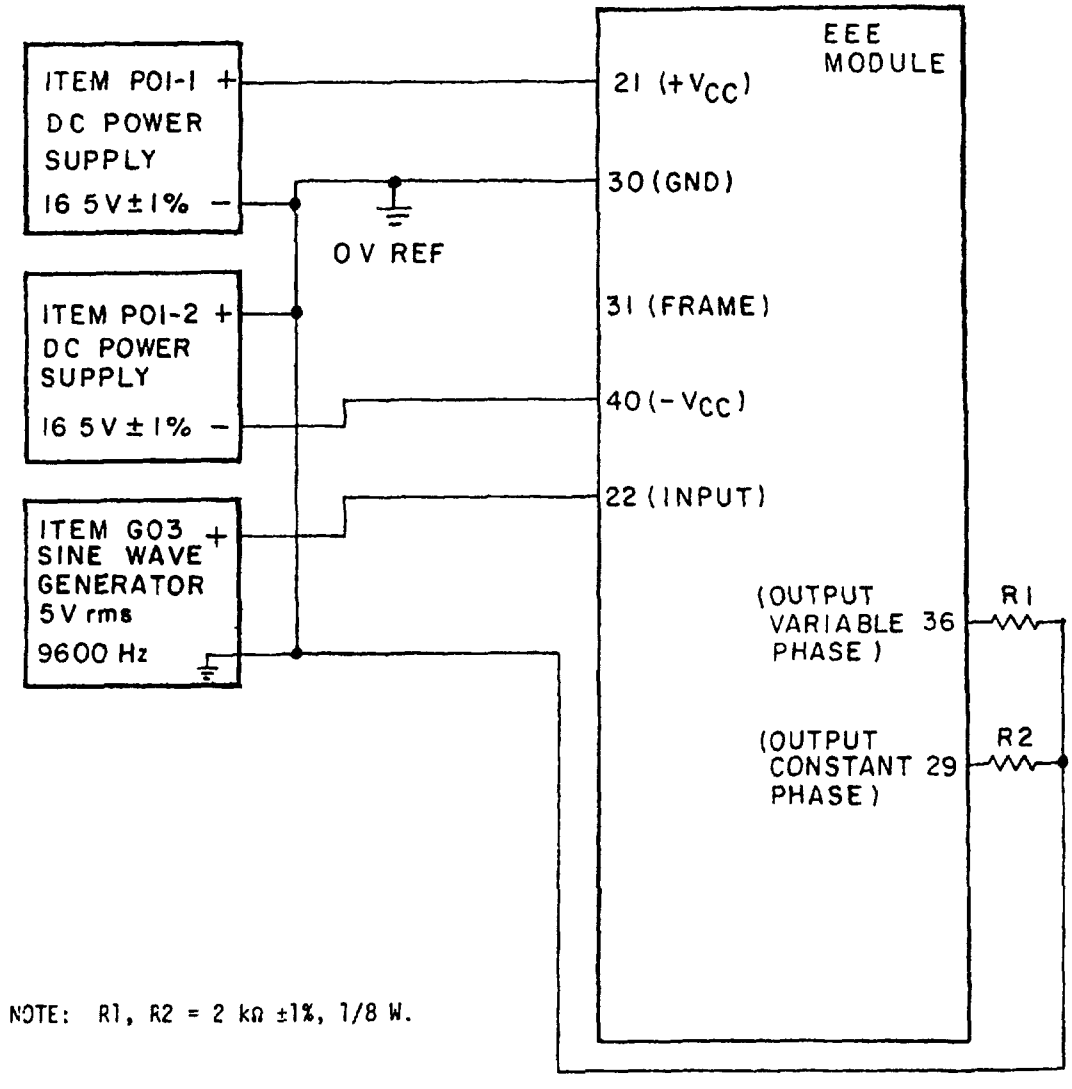
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NOTE: R1, R2 = 2 kΩ ±1%, 1/8 W.

FIGURE 3. Operating temperature, component temperature and life test circuit.

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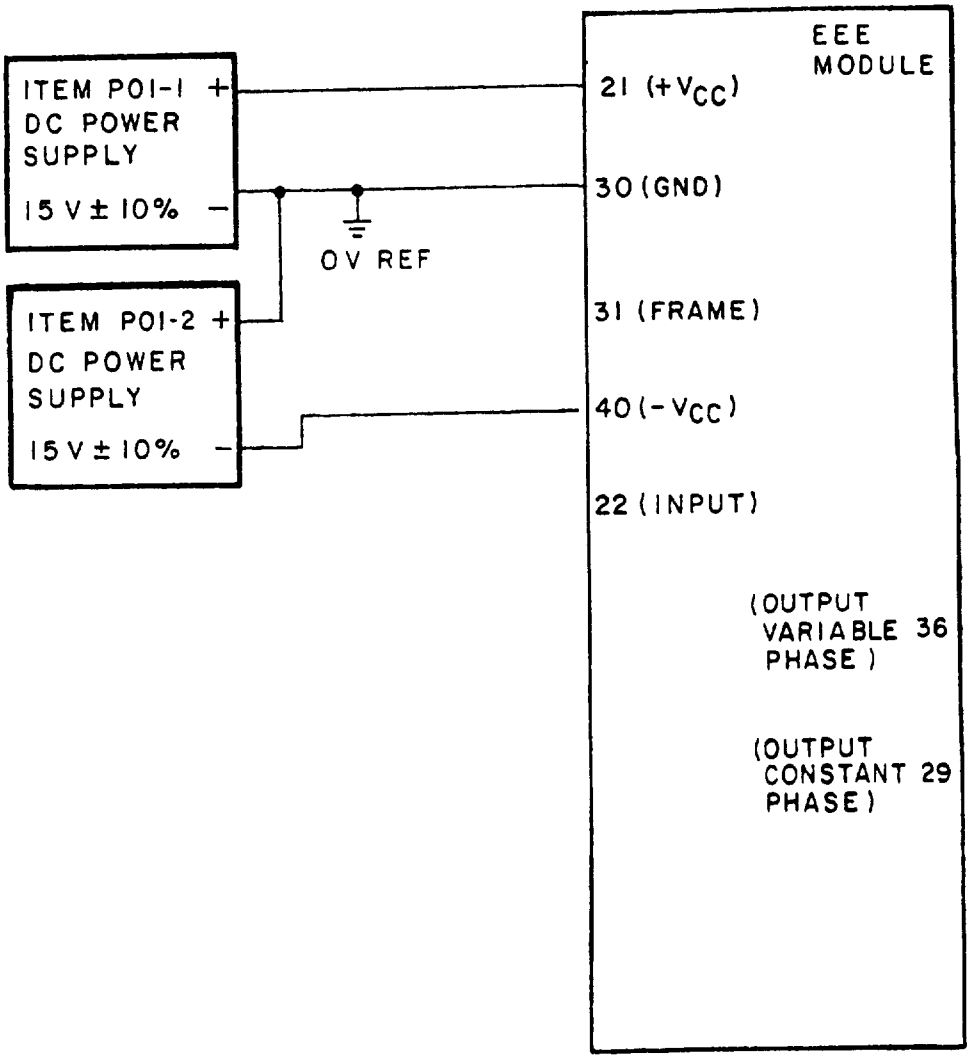


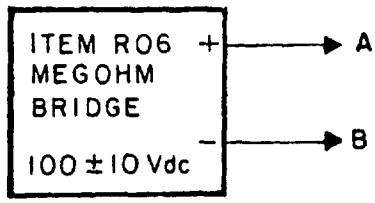
FIGURE 4 Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	21,22,30,31,29, 40,36	23,24,25,26,27, 28,32,33,34, 35,37,38,39, and 1 through 20 if supplied	10	∞	MΩ	5
A2	Isolation	31	21,22,23,24,25, 26,27,28,29,30, 32,33,34,35,36, 37,38,39,40, and 1 through 20 if supplied	10	∞	MΩ	5

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

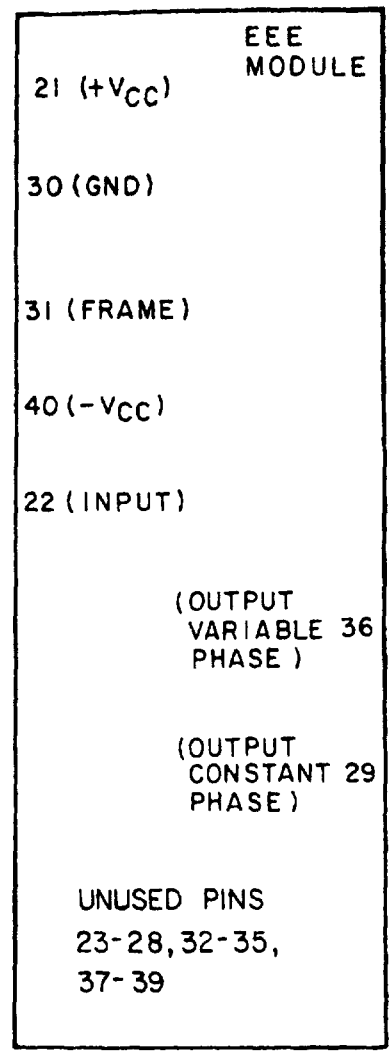


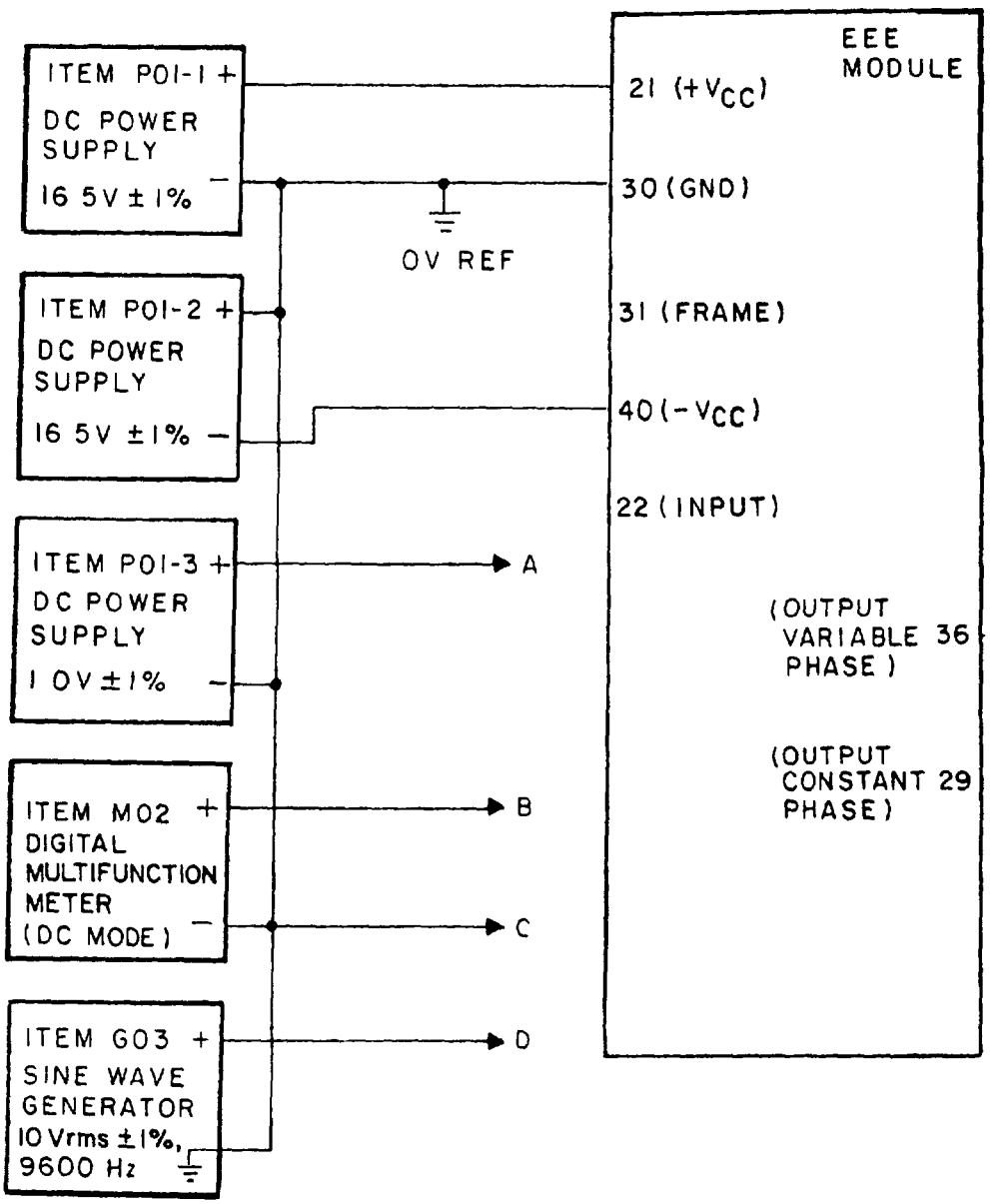
FIGURE 5 Isolation test circuit for table II

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TABLE III. Output short circuit duration, DC offset voltage, and DC gain test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
B1	Output short circuit duration	Open	Open	36	22	10	---	s	6
B2	Output short circuit duration	↓	Open	29	22	10	---	s	↓
C1	DC offset voltage	↓	36	22	Open	-20	20	mV	↓
C2	DC offset voltage	↓	29	22	↓	-15	15	mV	↓
D1	DC gain	22	36	Open	↓	0.95	1.05	V	↓
D2	DC gain	22	29	Open	↓	0.95	1.05	V	↓

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NOTES.
1. For test condition B1 - connect point C to pin 36 for at least 10 seconds
2. For test condition B2 - connect point C to pin 29 for at least 10 seconds

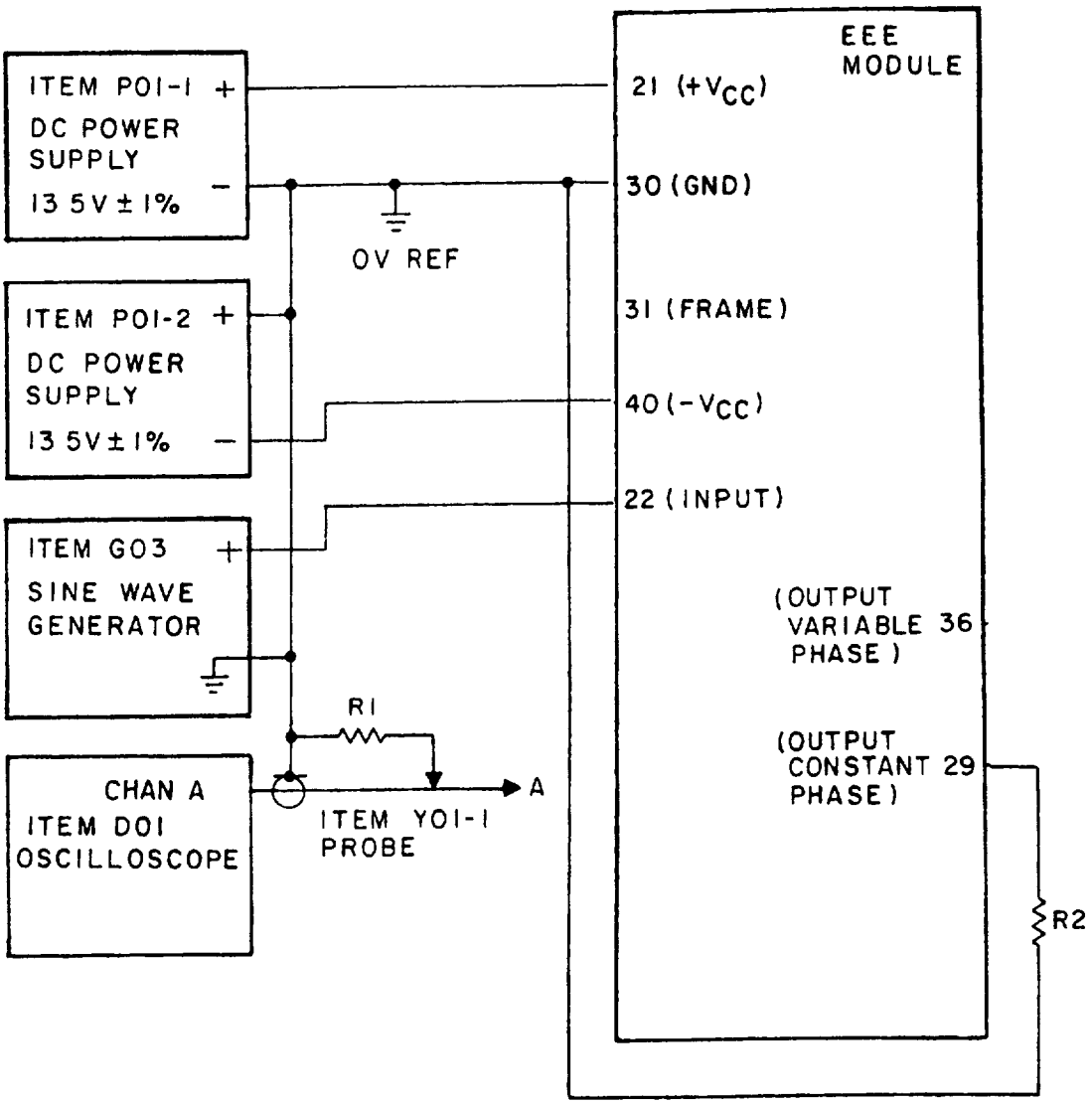
FIGURE 6. Output short circuit duration, DC offset voltage, and DC gain test circuit for table III.

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TABLE IV. Output voltage swing (positive and negative) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E1	Output voltage swing, positive (V_P)	36	7.0	---	V	7,8
E2	Output voltage swing, negative (V_N)	36	7.0	---	V	7,8

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NOTE R1, R2 = 2 kΩ ±1%, 1/8 W

FIGURE 7. Output voltage swing test circuit for table IV

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NOTES:
1. Input waveform from item G03.
   Frequency = 5000 Hz ±10%
   Voltage = Adjust the input voltage until the output waveform is clipped
2. When the output wave is clipped, measure V_p and V_N.

FIGURE 8. Output voltage swing test circuit waveform for table IV.

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TABLE V. Power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
F1	Power supply current $+V_{CC}$	1.0	8.5	mA	9
F2	Power supply current $-V_{CC}$	1.0	8.5	mA	9

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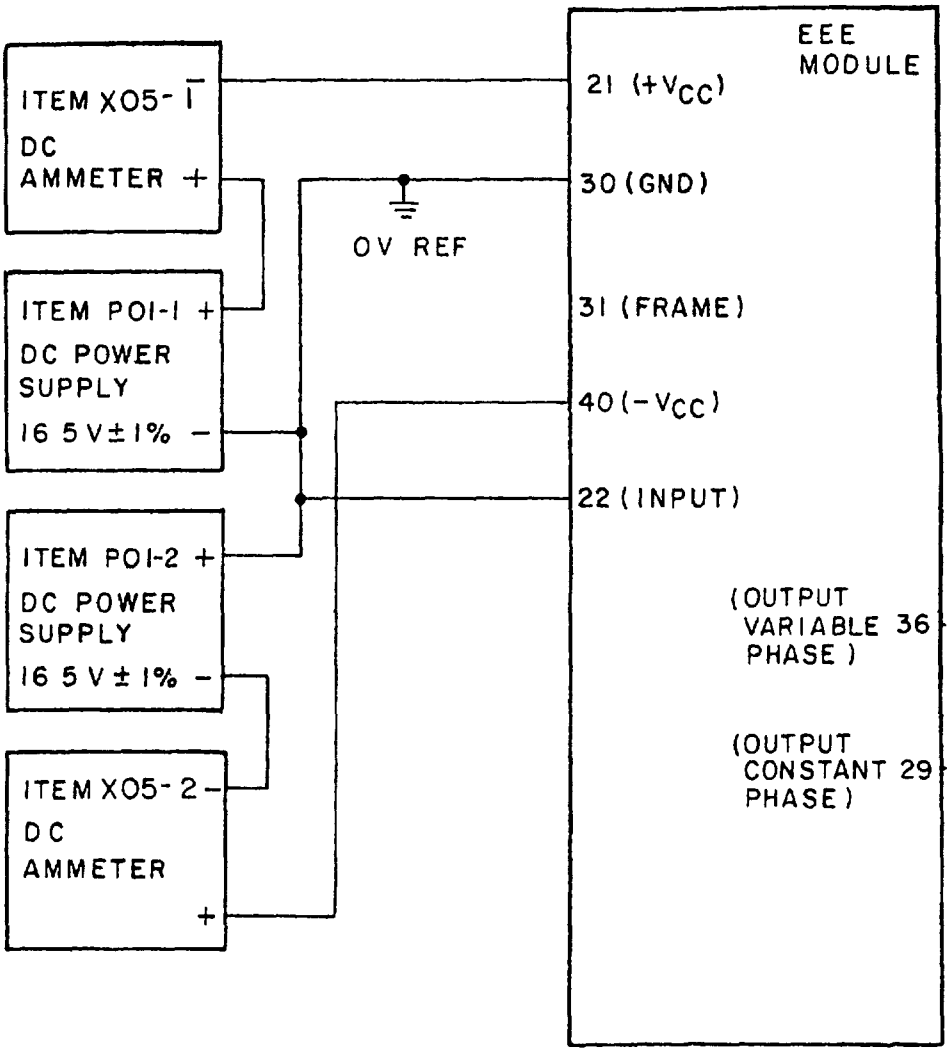


FIGURE 9 Power supply current test circuit for table V.

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TABLE VI. Gain versus frequency check test.

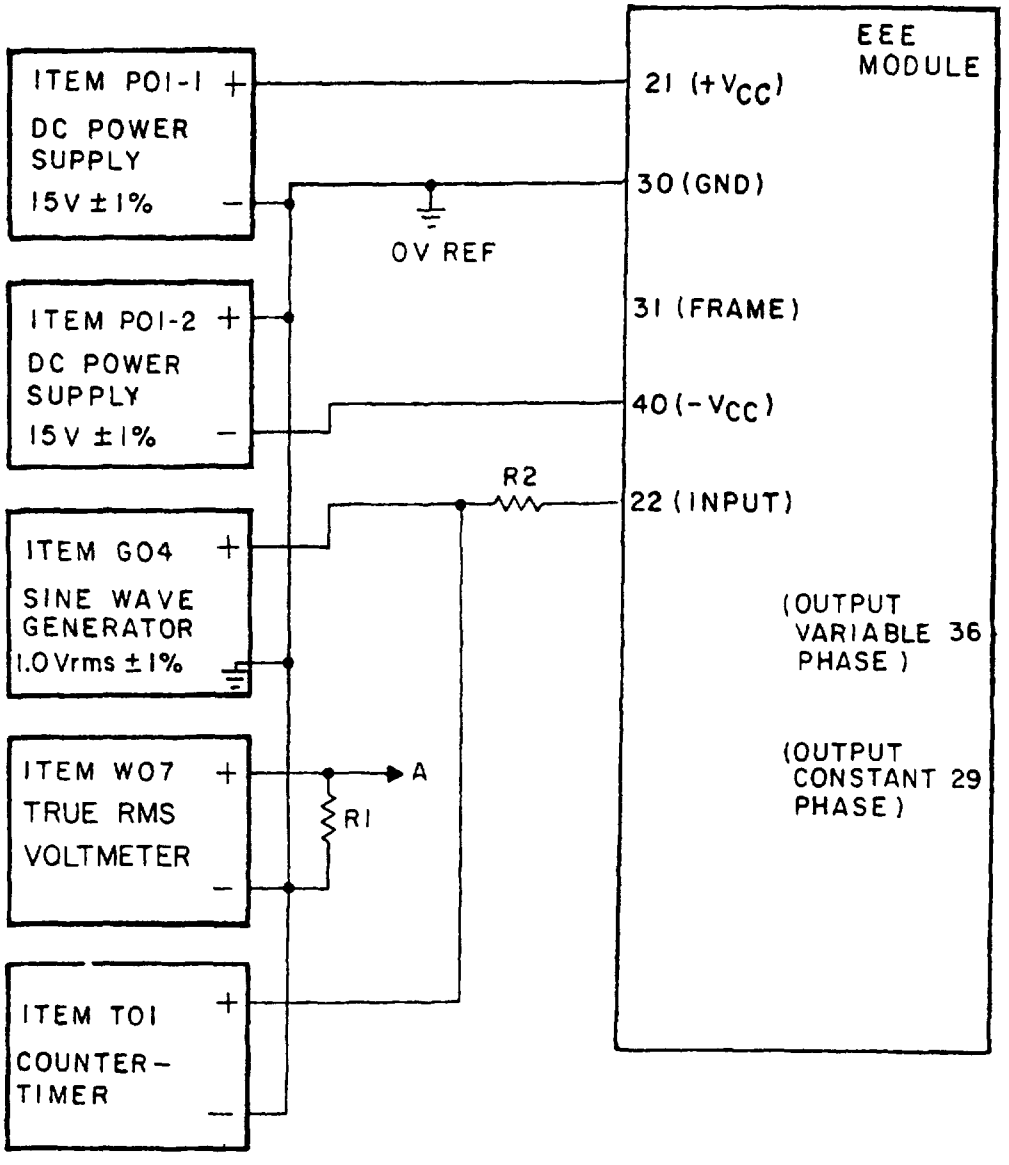
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G1	Gain versus frequency check, e_o at 9600 Hz	36	-0.5	1.5	dB	10
G2	Frequency sweep e_o at 20 to 7000 Hz (ref G1)	↓	-2.0	0.5	↓	↓
G3	Gain versus frequency check, e_o at 7600 Hz (ref G1)	↓	-0.3	0.3	↓	↓
G4	Gain versus frequency check, e_o at 9540 Hz (ref G1)	↓	-0.25	0.25	↓	↓
G5	Gain versus frequency check, e_o at 9660 Hz (ref G1)	↓	-0.25	0.25	↓	↓
G6	Gain versus frequency check, e_o at 11600 Hz (ref G1)	↓	-0.3	0.3	↓	↓
G7	Gain versus frequency check, e_o at 15 kHz (ref G1)	↓	-0.5	0.5	↓	↓
G8	Frequency at 3 dB down	↓	17.5	19.5	kHz	↓
G9	Gain versus frequency check, e_o at 50 kHz (ref G1)	↓	---	-40	dB	↓
G10	Frequency sweep e_o at 16 to 100 kHz	← See note 6, figure 9 →				
G11	Gain versus frequency check, e_o at 9600 Hz	29	-0.5	1.5	dB	↓
G12	Frequency sweep e_o at 20 to 7000 Hz (ref G11)	↓	-2.0	0.5	↓	↓
G13	Gain versus frequency check, e_o at 7600 Hz (ref G11)	↓	-0.3	0.3	↓	↓

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TABLE VI. Gain versus frequency check test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G14	Gain versus frequency check, e_o at 9540 Hz (ref G11)	29	-0.25	0.25	dB	10
G15	Gain versus frequency check, e_o at 9660 Hz (ref G11)		-0.25	0.25		
G16	Gain versus frequency check, e_o at 11600 Hz (ref G11)		-0.3	0.3		
G17	Gain versus frequency check, e_o at 15 kHz (ref G11)		-0.5	0.5		
G18	Frequency at 3 dB down		17.5	19.5	kHz	
G19	Gain versus frequency check, e_o at 50 kHz (ref G11)		---	-40	dB	
G20	Frequency sweep e_o at 16 to 100 kHz	See note 6, figure 9				

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NOTES
1. R1 = 2 kΩ ±1%, 1/8 W
2. R2 = 200Ω ±1%, 1/8 W.
3. Set the input from item G04 to 1.0 ±1% Vrms with all circuit connections made. Use 50Ω output on item G04.
4. Test condition G1, measure and record e_o at 9600 Hz.
5. Adjust all frequencies to ±1%
6. Test conditions G10 and G20, e_o decreases continuously with increasing frequency with no discernable peaking, that is, e_o at $f_1 > e_o$ at f_2 for $f_2 > f_1$.
7. Test conditions G8 and G18, adjust frequency until e_o is -3 dB, reference level measured in test condition G1 or G11 as applicable Record this frequency from item T01.

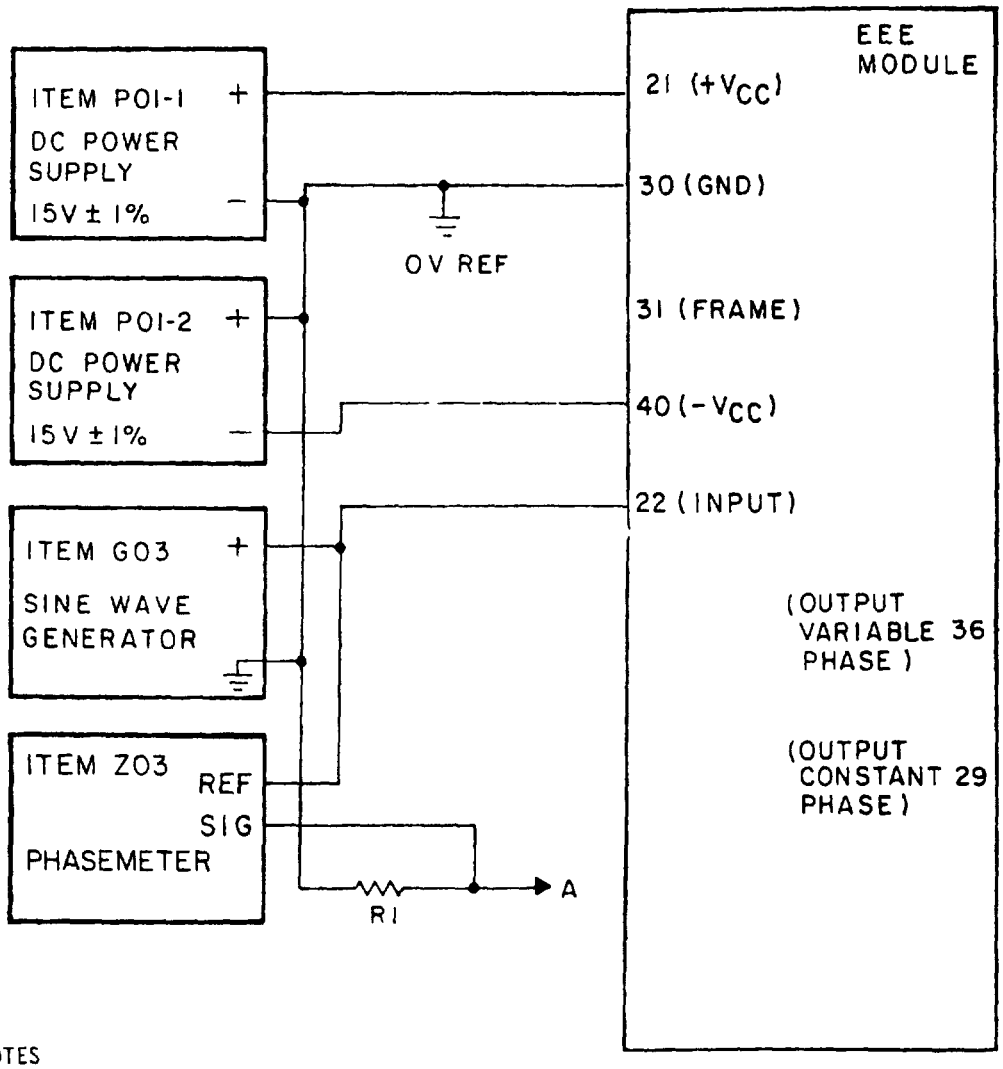
FIGURE 10. Gain versus frequency check test circuit for table VI.

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TABLE VII. Phase shift test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
H1	Phase shift (potentiometer fully clockwise)	36	240	260	Degree	11
H2	Phase shift (potentiometer, fully counterclockwise)	36	90	110		
H3	Phase shift	29	85	100		
H4	Phase shift (settability)	36	-0.5	0.5		
H5	Phase shift (setting under pressure)	---	-0.5	0.5		
H6	Phase shift (average temperature coefficient) check	---	---	---		

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NOTES
1. R1 = 2 kΩ ±1%, 1/8 W
2. Set item G03 output at 9600 Hz ±10% and 1.0 Vrms ±1%.
3. Test condition H4. Turn potentiometer cw until phasemeter reads 120° ±0.5°. Record this reading. Turn potentiometer 2 turns cw, then turn potentiometer ccw until phase is same as before. Maximum difference in reading shall be less than 0.5 degree.
4. Test condition H5. With potentiometer at last setting of condition H4, apply a pressure of 16 ounces (4.448 newtons) to potentiometer adjusting screw. Phase shall not change more than ±0.5 degree.
5. Test condition H6. For any potentiometer setting where phase shift is greater than 120 degrees, the average temperature coefficient of phase shift defined as

$$\frac{\Delta\phi}{\Delta T} = \frac{\text{Phase shift at } 0^{\circ}C - \text{phase shift at } 60^{\circ}C}{60^{\circ}C}$$

shall be less than $0.05^{\circ}/^{\circ}C$

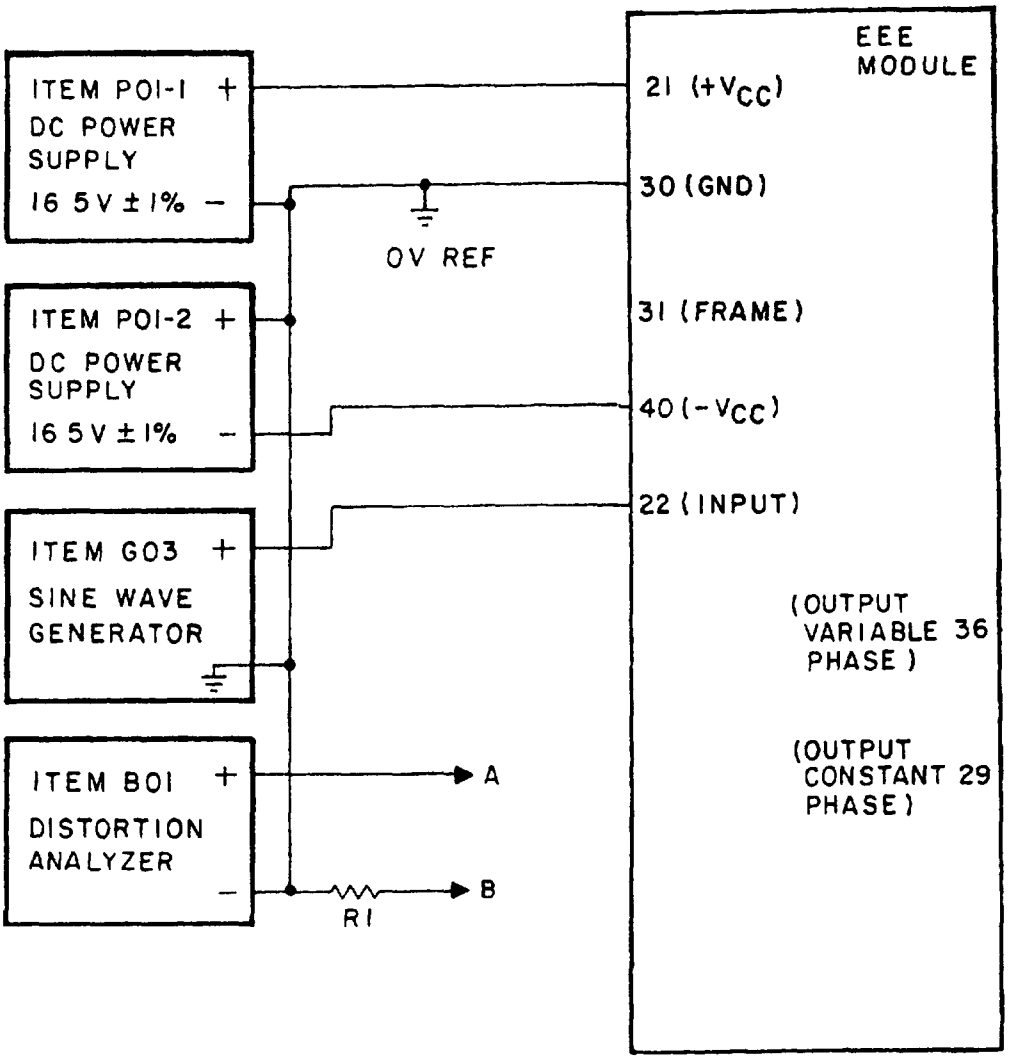
FIGURE 11. Phase shift test circuit for table VII.

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TABLE VIII. Distortion check test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		B to pin	Min	Max		
J1	Distortion check	36	0	2	%	12
J2	Distortion check	29	0	2	%	12

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NOTES.
1. R1 = 2 kΩ ±1%, 1/8 W.
2. Set the input from the item G03 to 4.0 Vrms ±1% at 9600 Hz ±10%.
3. Test condition J1 Connect point A to pin 22, and measure input distortion, D_1. Then connect point A to point B and measure output distortion, D_0. Then claculate distortion, D_1 with the formula:

$$D = D_0 - D_1$$

4. Repeat the above procedure for test condition J2

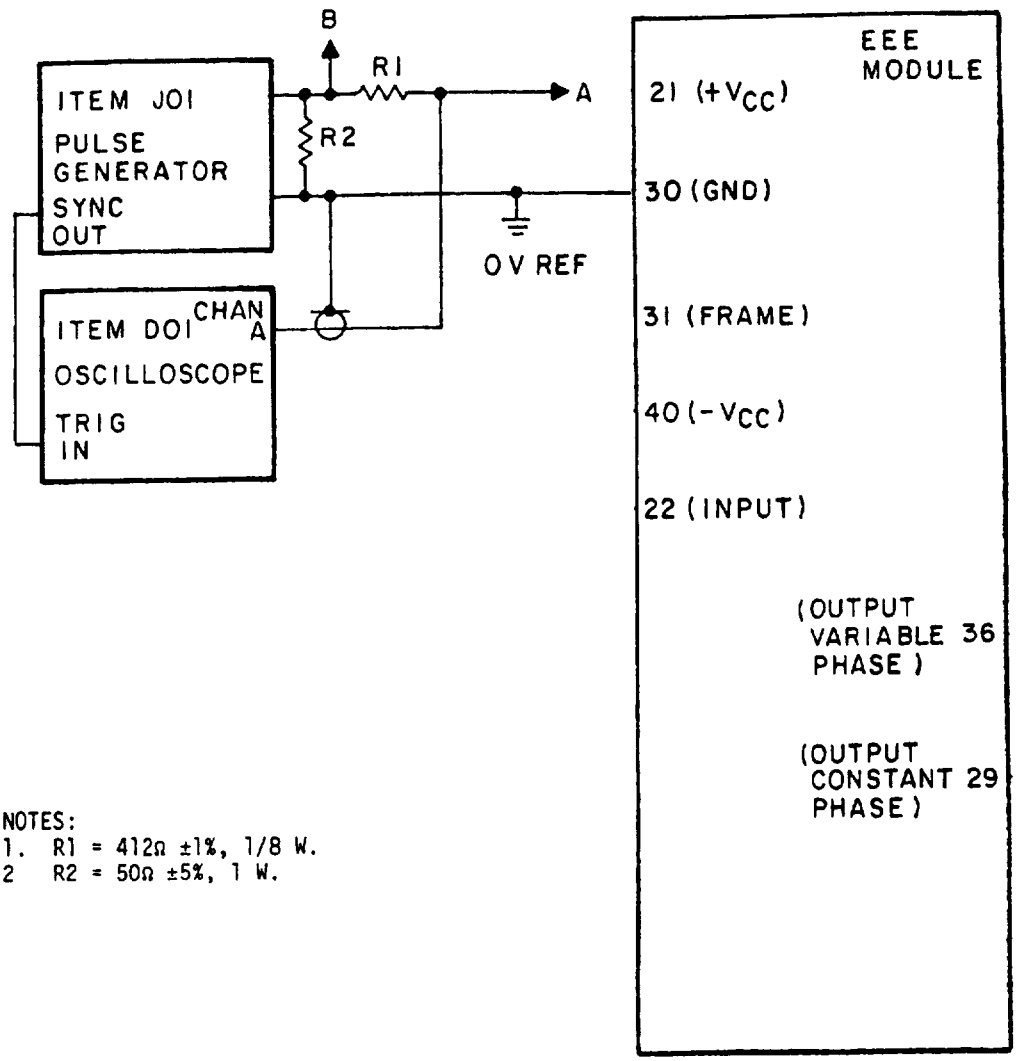
FIGURE 12. Distortion check test circuit for table VIII.

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TABLE IX. Power supply transient rise time test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
K1	Power supply transient rise time, $+V_{CC}$	21	27.5	77.5	µs	13,14
K2	Power supply transient rise time, $-V_{CC}$	40	27.5	77.5	µs	13,14

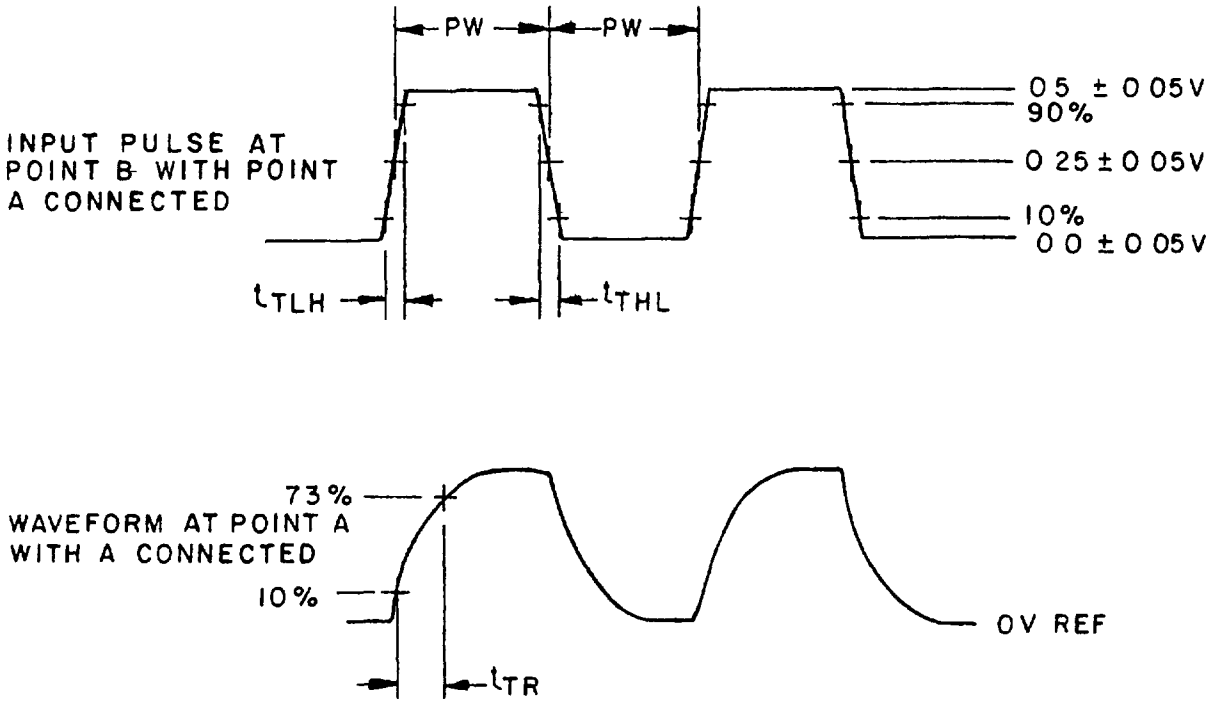
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NOTES:
1. R1 = 412Ω ±1%, 1/8 W.
2 R2 = 50Ω ±5%, 1 W.

FIGURE 13 Power supply transient rise time test circuit for table IX.

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NOTES
1. Input pulse parameters
   PW = 300 ±3 µs
   $t_{TLH} = t_{THL}$ = 0.50 ±0.05 µs

2. Measure t_{TR} at point A.

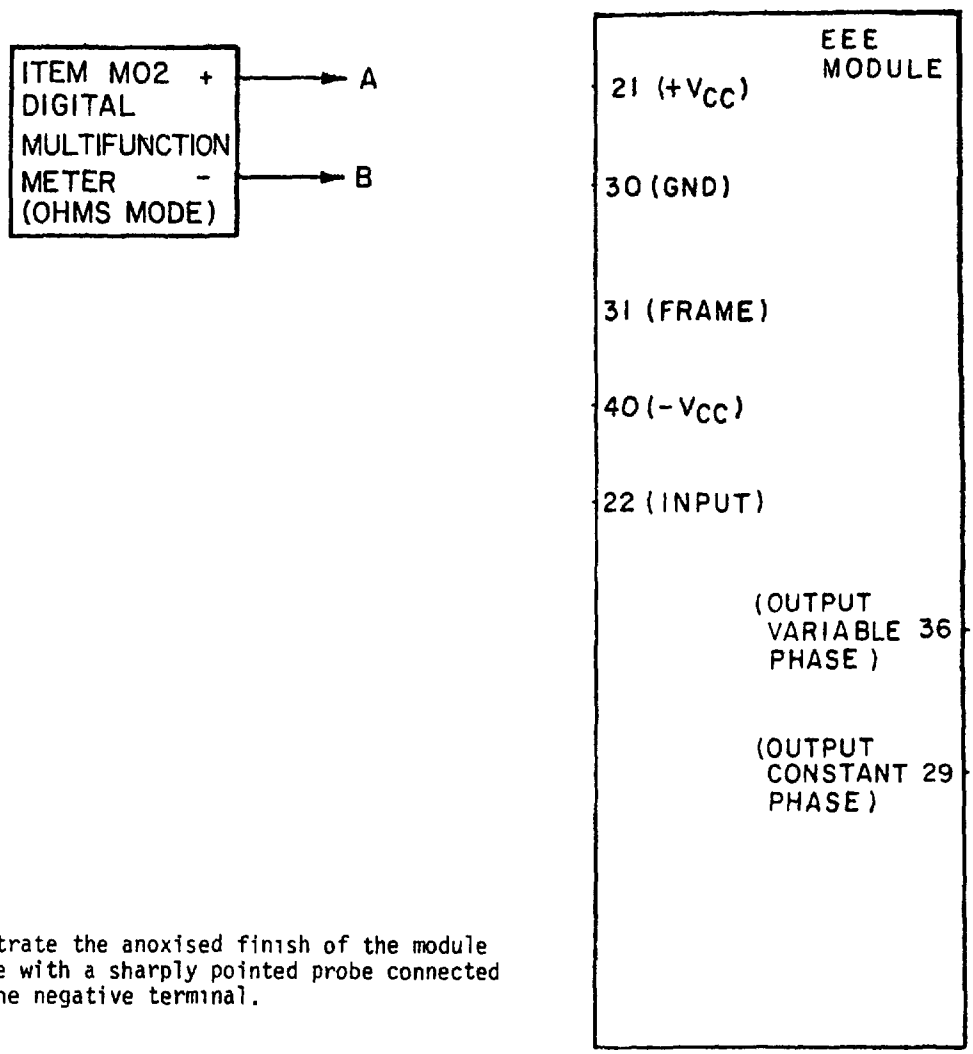
FIGURE 14. Power supply transient rise time test circuit waveforms for table IX.

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TABLE X. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L	Continuity	31	Frame	0	1	Ω	15

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NOTE Penetrate the anoxised finish of the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 15. Continuity test circuit for table X.

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